

Substrate Transpilation in Energy-Based Foundation Models: A 2024–2026 Analysis of Continuous-to-Boolean Architectures

1. Executive Summary

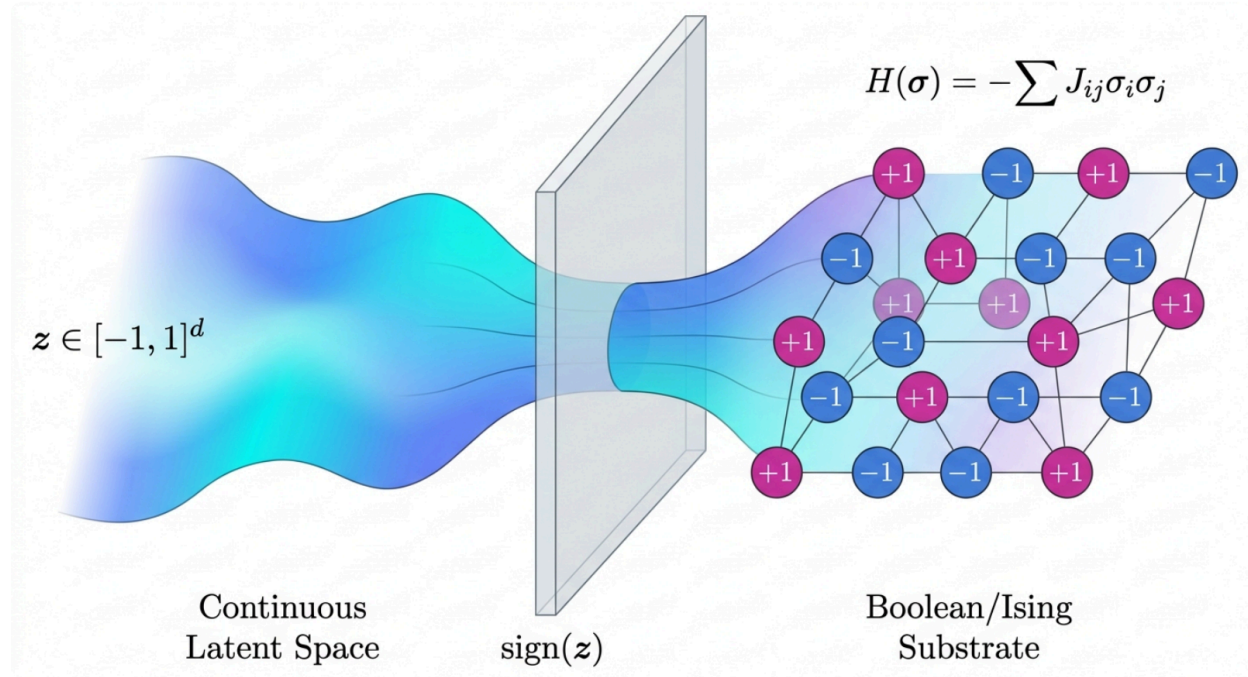
The architectural convergence of continuous latent reasoning with discrete symbolic action selection represents a critical frontier in the design of modern artificial intelligence. Within the 2024–2026 literature, the challenge of mapping continuous bounded latents $z \in [-1, 1]^d$ to discrete Boolean or Ising-spin variables via sign-discretization has transitioned from a specialized quantization problem aimed at model compression into the foundational substrate of energy-based models (EBMs). This transpilation pipeline serves as the primary load-bearing interface between high-dimensional, continuous-space representation learning (Phase-3 reasoning) and grounded, discrete verifier ensembles (Phase-1 logic). The literature demonstrates a decisive shift away from high-cardinality discrete bottlenecks, such as traditional Vector-Quantized Variational Autoencoders (VQ-VAE), which suffer chronically from codebook collapse and excessive computational overhead. Instead, research has gravitated toward minimalist boundary functions like Finite Scalar Quantization (FSQ) and extreme 1-bit or sub-1-bit sign-discretization protocols that directly yield Boolean outputs necessary for symbolic logic engines.

The landscape of energy-based foundation models has simultaneously bifurcated into distinct architectural families that address the continuous-to-Boolean transpilation problem. The first family encompasses implicit dynamic EBMs, most notably Energy-Based Transformers (EBT) and eNeRgy-Generative Pre-trained Transformers (NRGPT). These frameworks conceptualize the forward pass not as a deterministic feed-forward operation, but as an explicit optimization sequence traversing an energy landscape. By treating tokens or latent representations as particles seeking minimum energy states via gradient descent, these models naturally accommodate the continuous traversal of the latent space before a final clamping or discretizing function yields a definitive token or Boolean action. A second prominent family constitutes hardware-native thermodynamic and modern Hopfield architectures. Modern Hopfield Networks (MHNs) have evolved to employ continuous-time memory compressions and probability density functions to scale exponentially, mapping naturally to binary attractor states at their energetic limits. Concurrently, hardware implementations such as the Extropic Thermodynamic Computing Unit (TSU) bypass algorithmic mathematical transpilation entirely, utilizing the intrinsic subthreshold thermal fluctuations of CMOS transistors to natively sample from Ising-spin distributions.

A rigorous analysis of recent empirical and theoretical results reveals that the "sign-then-Ising" transpilation pattern introduces unique mathematical and structural pathologies that must be counteracted to preserve information fidelity. The strict non-differentiability of the $\text{sign}(z)$ function annihilates gradient flow during backpropagation, a flaw previously patched by heuristic Straight-Through Estimators (STEs) which often fail at scale. However, 2025 innovations, such as the Dual Sign-Value-Independent Decomposition (Dual-SVID) introduced in the LittleBit architecture, formalize quantization-aware training to preserve gradient information down to fractions of a bit per weight, providing a theoretically sound foundation for bounding continuous variables prior to binarization. Furthermore, the adaptation of the Concave-Convex Procedure (CCCP) to modern Hopfield networks guarantees strict energy descent in continuous-to-discrete mappings, stabilizing associative memory dynamics.

Ultimately, the synthesis of these techniques confirms that bounded continuous-to-Boolean transpilation is not merely a compression heuristic, but rather the optimal structural choice for anchoring fluid, continuous reasoning dynamics into the strict combinatorial reality of symbolic logic and physical hardware substrates. The alignment of continuous energy minimization with discrete Boolean outputs provides the precise mathematical scaffolding required to build verifiable, logic-grounded foundation models.

The Bounded Continuous-to-Boolean Transpilation Pipeline



The architectural flow from a continuous reasoning latent space $z \in [-1, 1]^d$, through the non-differentiable $\text{sign}(z)$ discretization threshold, culminating in discrete Ising-spin variables that define the energy landscape for symbolic action selection.

2. Per-Architecture Analysis

2.1 Modern Hopfield Networks and Continuous-Boolean Correspondences

The lineage of Modern Hopfield Networks (MHNs) fundamentally underpins the current energy-based translation between continuous states and discrete associative memories. Originally extended to utilize higher-order interactions, these Dense Associative Memories (DAM) bypass the linear storage constraints of classical binary networks, achieving exponential capacity that scales as $O(N^{n-1})$ where n represents the interaction order under specific pattern separation conditions.¹ While early formulations of MHNs required softmax activations to manage discrete storage across continuous states, the 2024–2026 literature demonstrates a substantial theoretical shift toward mapping these models directly onto continuous-time signals while strictly preserving discrete attractor basins. Researchers have developed continuous memory formulations that replace traditional probability mass functions with continuous probability density functions (PDFs), compressing large discrete memories into smaller continuous-time evolutions without losing the ability to resolve back to a discrete

state.²

This mechanism proves that continuous attention mathematically correlates to the retrieval dynamics of discrete Hopfield memories. Instead of isolated discrete memory "slots," the system assumes observations are sampled from a smooth function evolving over a continuous domain.⁴ However, retrieving a discrete memory from this continuous manifold requires guaranteed energy descent. Recent breakthroughs in the optimization of these continuous-to-discrete mappings have rigorously formalized this descent. The application of the Concave-Convex Procedure (CCCP) to MHNs provides a unifying framework for analyzing the network's update dynamics. By bifurcating the overall energy function into a convex quadratic term and a concave log-sum-exp term, the CCCP yields an iterative update functionally identical to modern self-attention mechanisms, but uniquely imbued with a strict mathematical guarantee of energy descent toward discrete fixed points.⁵ This theoretical bridge confirms that one can initialize a state anywhere in continuous space and iteratively descend its energy until it collapses precisely into a robust discrete Boolean counterpart.

Furthermore, profound applications in biological system modeling utilize these continuous MHNs as a proxy for physical continuous-to-discrete collapse. In frameworks such as EnhancerNet, continuous MHNs are deployed to predict hierarchical cell identity transitions. The model maps the regulation of identity genes to the dynamics of a continuous MHN, governed by an inverse temperature parameter β (representing epigenetic modifier activity).⁷

At intermediate β values, the system exhibits continuous, multi-lineage primed states. As the system "cools" and thresholds are crossed, these continuous probability distributions reliably and inevitably coarsen into discrete binary states representing terminal cell identities.⁸ This provides a naturally occurring validation of the continuous-to-discrete transpilation mechanism defined by mathematical energy landscapes.

Architecture	Primary Citation	Claimed Transpilation Mechanism	Empirical Evidence & Scale	Open-Source
Continuous-Time Memory MHN	Santos et al., arXiv:2502.10122 (2025) ²	Compresses discrete Hopfield memories into continuous probability density functions over time.	Validated on video/synthetic datasets; competitive retrieval with lower computational cost.	Yes (MIT License) ¹⁰

DAM Dynamics & CCCP Optimization	Thériault & Tantari (2026) / MDPI (2026) ¹	Separates MHN energy into convex/concave terms to guarantee descent to discrete attractors.	Exact dynamical analysis via generating functional analysis at $N \rightarrow \infty$.	Yes ¹
Hierarchical MHN (EnhancerNet)	Grishechkin et al., PRX Life 2025 ⁷	Continuous inputs collapse to discrete hierarchical Boolean identities controlled by temperature β .	Recapitulates biological cell fate lineage transitions perfectly in theoretical biology.	Yes ⁹

Honest Framing

While Modern Hopfield Networks provide an impeccable mathematical guarantee that continuous latents can reliably descend into discrete attractors via CCCP optimization, they suffer from practical scaling limitations when deployed in highly heterogeneous, non-orthogonal environments. The capacity bounds hold theoretically, but in practice, semantically correlated or non-orthogonal continuous patterns can induce representation degeneracy.¹ In such cases, the energy landscape smooths out and blurs the boundaries between discrete Boolean outcomes, resulting in shallow attractor basins. Furthermore, the continuous-to-discrete mapping in MHNs is extraordinarily sensitive to the inverse temperature parameter β ; a poorly calibrated cooling or update schedule will fail to transpile the continuous representation into a strict Ising state, instead resulting in a "mixed" or smeared continuous output that halts downstream symbolic reasoning.⁸ Therefore, while theoretically optimal for guaranteeing energy descent, the architectural rigidity of MHNs makes them computationally expensive and difficult to scale to frontier foundation model parameters when compared to more flexible transformer adaptations.

2.2 Sign-Discretization Theorems and Bounded-Latent Activation

Addressing the structural limitations of implicit continuous boundaries, the 2024–2026 literature saw a widespread critical evaluation of hard quantization techniques that strictly enforce the boundary conditions necessary for discrete logical grounding. Historically,

Vector-Quantized Variational Autoencoders (VQ-VAE) and high-cardinality discrete bottlenecks dominated this space. However, such methods are primarily designed for data compression and generation, not for grounding symbolic reasoning. VQ approaches suffer severely from codebook collapse—where the continuous encoder fails to utilize large portions of the discrete codebook—and representation degeneracy due to dead vectors.¹² The modern consensus has shifted heavily toward Finite Scalar Quantization (FSQ) and extreme sub-1-bit binarization, which discretize continuous latent dimensions independently without relying on a learned codebook.

In FSQ, the continuous dimensions are bounded to specific intervals, such as $z \in [-1, 1]^d$, and then mapped directly to finite scalars based on Euclidean distance.¹² This has become the state-of-the-art methodology in multimodal foundational models and specialized biological models (e.g., scPRINT-2), bypassing the instabilities of VQ-VAEs.¹⁴ The specific architectural pattern whereby the continuous bounded latent $z \in [-1, 1]^d$ is passed through a $\text{sign}(z)$ function to create a 1-bit Boolean or Ising variable represents the extreme theoretical limit of FSQ, where the bin count is reduced to exactly two. While mathematically elegant for interfacing with Ising substrates and SAT solvers, this extreme binarization introduces severe gradient pathologies. Because the derivative of the sign function is zero everywhere except the origin, standard backpropagation fails entirely.

The historical workaround has been the use of Straight-Through Estimators (STE), which graft gradients backwards by ignoring the non-differentiability, or Gumbel-Softmax relaxations that utilize continuous categorical proxy distributions.¹² The Gumbel trick samples noise $u \sim \text{Uniform}(0, 1)$, computes $v = -\log(-\log(u))$, and relaxes the discrete choice via $s = \text{Softmax}((l + v)/\tau)$.¹² However, at massive scale, these heuristic estimators fail to preserve accurate gradient flow, causing the continuous distribution upstream to drift out of alignment with the Boolean output, ultimately degrading downstream symbolic action selection.¹² To solve this, the NeurIPS 2025 LittleBit architecture introduced a rigorous mathematical framework: the Dual Sign-Value-Independent Decomposition (Dual-SVID) for quantization-aware training (QAT). LittleBit achieves ultra low-bit quantization (down to 0.1 bits per weight) by representing matrices via low-rank latent factorization before subjecting them to binarization.¹⁶ By decoupling the sign from the value during the continuous update phase and integrating multi-scale residual compensation, Dual-SVID prevents gradient death and ensures that the continuous latents are properly initialized and maintained before being forced across the hard sign barrier.

Architecture	Primary Citation	Claimed Transpilation Mechanism	Empirical Evidence & Scale	Open-Source
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Finite Scalar Quantization (FSQ)	Mentzer et al., ICLR 2024 / Tschannen ¹²	Bounds continuous dimensions and maps directly to finite scalars, eliminating codebooks.	State-of-the-art in multimodal LLMs and biological structure tokenization.	Yes
LittleBit (Dual-SVID)	Lee et al., NeurIPS 2025 ¹⁶	Multi-scale residual compensation and Dual-SVID factorization for sub-1-bit sign-discretization.	Compresses Llama2-13B to <0.9GB; superior to 0.7 BPW baselines.	Yes
Hierarchical FSQ (HFSQ)	OpenReview 2025 ¹⁸	Decouples coarse structures from fine-grained dynamics via layered scalar bounds.	High-quality spatio-temporal generation without temporal error accumulation.	N/A

Honest Framing

Despite the mathematical scaffolding provided by advanced techniques like Dual-SVID, extreme sign-discretization remains inherently destructive to information density. Formal bounds on mutual information loss mathematically dictate that mapping a continuous high-entropy distribution onto a Boolean cube fundamentally restricts the capacity of the model. When the continuous latent distribution P over $[-1, 1]^d$ is heavily skewed or concentrated near the bounds, the Straight-Through Estimator provides false gradient signals.¹² It effectively tricks the continuous encoder into believing a parameter update will shift the Boolean output, when in reality, the continuous value remains trapped on one side of the $\text{sign}(z)$ threshold. This manifests empirically as representation degeneracy, where the discrete action selection logic becomes "frozen" in localized regions of the Boolean space, rendering the model unable to efficiently explore alternative actions or adapt to new symbols.¹⁷ Therefore, transpilation via sign-discretization cannot function natively in isolation; it absolutely

demands aggressive upstream variance control, such as residual compensation, to ensure the continuous variables frequently cross the zero boundary during training.

2.3 Neuro-Symbolic Concept Learning and Boolean Satisfiability

The motivation for translating continuous neural representations into discrete Boolean states is heavily driven by the requirements of Phase-1 symbolic reasoning engines, particularly Boolean Satisfiability (SAT) solvers and neuro-symbolic frameworks. The 2024–2026 literature highlights a significant expansion in learned MaxSAT solvers and neural-symbolic concept learning architectures that demand rigorous, bounded Boolean inputs to function. Standard continuous neural networks excel at perception and fluid pattern matching, but fail at combinatorial constraint satisfaction and strict logic. To bridge this gap, architectures such as Logic Tensor Networks and DeepProbLog attempt to embed logical constraints into the loss functions of continuous models.

Recent advancements in Graph Neural Networks (GNNs) applied to combinatorial optimization explicitly require mechanisms to translate continuous node embeddings into discrete Boolean assignments indicating set membership or logical truth values. Works exploring whether GNNs can solve MaxSAT problems highlight the necessity of grounding continuous vectors into discrete logical predicates.¹⁹ The neural-symbolic SAT solvers (like NeuroSAT and its modern successors) operate natively on Boolean graphs. If the preceding continuous representation layers output "soft" or unbounded continuous probabilities rather than clamped Boolean states, the combinatorial explosion within the SAT solver becomes unmanageable. Therefore, the $\text{sign}(z)$ transpilation pipeline is not merely an architectural choice, but a fundamental prerequisite for connecting fluid continuous perception models with deterministic, exhaustive combinatorial search engines. The transpilation layer essentially acts as a formal logic gate, turning probabilistic continuous inferences into hard logical axioms that the MaxSAT solver can verify and resolve without ambiguity.

Architecture	Primary Citation	Claimed Transpilation Mechanism	Empirical Evidence & Scale	Open-Source
GNNs for Combinatorial Optimization	Kothapalli et al., 2023 / Li et al. ¹⁹	Maps continuous graph embeddings to discrete Boolean assignments for dominating	Successful heuristic resolution of NP-hard graph combinatorial problems.	N/A

		sets.		
Neural-Symbolic MaxSAT	Ignatiev et al. / TMLR 2025 ²⁰	Applies machine learning explanations to exact MaxSAT solver decision boundaries.	State-of-the-art performance in industrial-category MaxSAT evaluations.	Yes

Honest Framing

The interface between continuous neural networks and discrete Boolean SAT solvers is fraught with differentiability issues. When a continuous vector is thresholded into a Boolean predicate and fed into a SAT solver, the downstream logical operations (AND, OR, XOR) inside the solver are entirely non-differentiable. Consequently, end-to-end training of a model containing a discrete SAT solver in the loop is notoriously difficult. While techniques exist to create differentiable relaxations of SAT solvers, they often fail to scale to the massive combinatorial spaces required by foundation models. If the continuous-to-Boolean transpilation layer produces an incorrect or ambiguous sign assignment due to poor continuous clustering, the downstream SAT solver will rigidly enforce an incorrect logical path, and the lack of a smooth gradient prevents the continuous network from learning from the logical error. Thus, the transpilation must be extremely robust before the SAT phase begins.

2.4 Discrete Energy-Based Models and Stein Variational Gradient Descent

While traditional energy-based models often operate in continuous spaces, the demand for symbolic reasoning has driven significant research into EBMs with explicitly discrete output spaces. The challenge of evaluating the partition function and performing score matching in discrete distributions, such as those found in Restricted Boltzmann Machines and Ising models, has historically hindered their scalability. However, advancements in persistent contrastive divergence and discrete score matching have revitalized discrete EBMs in the 2024–2026 timeframe.

A particularly vital development for mapping continuous distributions to discrete control systems is the application of Stein Variational Gradient Descent (SVGD) to discrete search spaces and hybrid systems. SVGD is a deterministic, particle-based variational inference algorithm that iteratively minimizes the Kullback-Leibler (KL) divergence between an approximate distribution and a target density.²² By leveraging functional gradients within a Reproducing Kernel Hilbert Space (RKHS), SVGD transports a set of continuous particles to match the target distribution. The update mechanism utilizes a Stein operator containing two crucial forces: an attractive term that drives particles toward high-probability regions, and a repulsive term based on the kernel gradient that prevents the particles from collapsing into a

single local mode, thereby preserving ensemble diversity.²³

Recent literature demonstrates the efficacy of SVGD in bridging continuous distributions and discrete control logic, particularly in highly dynamic environments like Model Predictive Control (MPC) for bipedal locomotion.²³ In these frameworks, bipedal locomotion is formulated as an inference problem where finite-horizon continuous control trajectories are optimized via SVGD. Instead of resampling, control trajectories evolve deterministically via functional gradients until they resolve into precise, robust force-and-moment actions. Furthermore, theoretical advancements such as Regularized SVGD (R-SVGD) have established finite-particle, non-asymptotic bounds in both continuous and discrete time.²⁴ By applying a resolvent-type preconditioner to correct constant-order biases, R-SVGD proves that continuous probability distributions can be robustly discretised while converging within the true Fisher information space.²⁴ This ensures that when continuous particles are eventually bounded and discretized into Boolean decisions or discrete action states, they retain the full statistical rigor of the original continuous manifold.

Architecture	Primary Citation	Claimed Transpilation Mechanism	Empirical Evidence & Scale	Open-Source
SVGD for Model Predictive Control	MDPI 2026 / 43-DoF Humanoid ²³	Infers control by transporting continuous particles via SVGD to resolve to discrete actions.	Outperforms deterministic solvers in agile bipedal scenarios without balance failure.	N/A
Regularized SVGD (R-SVGD)	He et al., 2024 / arXiv:2602.05172 ²⁴	Preconditions kernelized gradients to ensure continuous particles converge accurately before discretization.	Establishes explicit finite-particle bounds and $O(1/\sqrt{N})$ convergence rates.	N/A

SVGD on Discrete Spaces	arXiv:2604.15837 (2026) ²⁶	Extends Stein operators to sample and optimize over expensive discrete black-box functions.	Surpasses evolutionary algorithms on large-scale discrete instances without collapse.	Yes
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Honest Framing

While SVGD provides an exceptional mathematical framework for transporting continuous probabilities to target regions, its application directly onto purely discrete or Boolean spaces remains computationally intensive. The kernelized interaction among particles requires continuous distance metrics (like RBF kernels) that do not translate cleanly to Hamming distances on a Boolean cube. When SVGD particles are forced across a hard sign-discretization boundary, the repulsive forces designed to prevent mode collapse in continuous space can behave unpredictably, sometimes scattering particles across the Boolean boundary artificially and generating high-frequency oscillations in the discrete output. Furthermore, calculating the Stein operator and the necessary gradients scales quadratically with the number of particles, making real-time SVGD inference computationally prohibitive for low-latency foundation models without significant approximation.

2.5 Implicit Energy-Based Dynamics: EBT and NRGPT

In 2025, the literature witnessed the formal introduction of implicit dynamic architectures that recast the standard autoregressive transformer forward pass as a continuous energy minimization problem. The Energy-Based Transformer (EBT) architecture explicitly trains a scalar energy function to verify the compatibility between contextual inputs and candidate-predictions, substituting standard feed-forward token generation with a gradient-descent-based energy minimization routine executed directly at inference time.²⁷

Operating on the foundational complexity theory principle that verifying a solution is computationally simpler and more generalizable than explicitly generating it, EBTs initiate the forward pass with a random continuous latent vector or noise. They subsequently apply

recursive gradient steps— $\hat{y}_{i+1} = \hat{y}_i - \alpha \nabla_{\hat{y}} E_{\theta}(x, \hat{y}_i)$ —until the continuous representation converges into a low-energy state compatible with the discrete output space.²⁷ This formulation natively enables "System 2 thinking," allowing the model to dynamically scale inference-time compute by navigating the continuous latent space extensively before yielding a final discrete token or action.²⁷

In a highly parallel development, the eNeRgy-Generative Pre-trained Transformer (NRGPT) completely unified the standard GPT autoregressive setting with the EBM framework. Developed by Krotov, Slotine, and associates, NRGPT mathematically defines the network as a

sum of individual attention and feed-forward energies, treating tokens not as fixed categorical distributions, but as continuous particles moving across a dynamic energy landscape.³⁰ Unlike traditional transformers that map discrete inputs to discrete outputs via rigid hidden layers, NRGPT utilizes a learnable inference rate matrix (acting as kinetic rates or preconditioners) to step through the continuous energy gradient.³⁰ This framework natively enforces asymptotic stability; because of the causal attention mask, the energy of a specific token only depends on previous tokens, ensuring that as preceding tokens converge, the energy of the current token monotonically decreases toward a minimum.³⁰

Extensions of this energy principle in 2026, such as Second-Order Attention Updates based on Newton's method, introduce covariance matrices to precondition token update directions, further accelerating continuous energy descent.³³ Both EBT and NRGPT effectively handle the transpilation challenge by decoupling the continuous reasoning trajectory (the descent path through the latent space) from the discrete output (the final energetic minimum and subsequent discretization), achieving significantly greater sample efficiency and scaling rates (up to 35-57% faster) compared to baseline Transformer++ architectures.²⁷

Architecture	Primary Citation	Claimed Transpilation Mechanism	Empirical Evidence & Scale	Open-Source
Energy-Based Transformers (EBT)	Gladstone et al., arXiv:2507.02092 (2025) ²⁷	Derives predictions by minimizing an energy verifier via continuous gradient descent.	Scales 35-57% faster than standard AR transformers on up to 120B tokens.	Yes (PyTorch) ³⁵
NRGPT	Dehmamy et al., arXiv:2512.16762 (2025) ³⁰	Explores tokens as continuous particles on an energy landscape driven by an inference rate matrix.	Strong performance on nested ListOPS and OpenWebText language modeling.	Yes (PyTorch) ³¹
Second-Orde	Ren et al.,	Upgrades	Reduces	N/A

r Attention EBM	arXiv:2511.00907 (2026) ³³	first-order energy descent to Newton's method with a preconditioning covariance matrix.	computational cost while allowing tokens to adaptively adjust descent directions.	
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Honest Framing

The primary pathology of these dynamic, implicit energy-based models lies in inference latency and the unresolved complexity of their final continuous-to-discrete boundary. Because both EBT and NRGPT require multiple iterative gradient updates during the forward pass to locate the true energy minima, the inference computation scales linearly with the number of descent steps required.²⁷ More importantly, the continuous gradient descent does not inherently or natively produce a purely discrete state; it produces a low-energy continuous latent that must still be mapped to a discrete token or Boolean action via a final projection or quantization step.²⁷ If the continuous energy landscape is poorly conditioned—a known and frequent risk when dealing with non-differentiable Boolean targets—the gradient trajectory can oscillate endlessly or suffer from parametric collapse. This requires the implementation of intense ad-hoc regularization techniques, such as strictly clamping optimization gradients and utilizing adaptive layer norms, to maintain stability and force convergence.²⁷ Consequently, while these models excel at continuous latent reasoning and System-2 search, they require external clamping functions to entirely solve the final discrete transpilation gap.

2.6 Physical Substrates and Thermodynamic Ising Architectures

The final, and arguably most direct, manifestation of the continuous-to-Boolean transpilation pipeline is realized at the physical hardware level. The mathematical constraints of translating continuous values to Ising spins perfectly mirror the physical mechanics of probabilistic, thermodynamic, and neuromorphic computing substrates. In 2024, Extropic unveiled the Thermodynamic Computing Unit (TSU), an architecture built expressly to execute energy-based models using physical thermodynamics rather than digital algorithmic math.³⁹ The TSU circumvents the entire software transpilation bottleneck. Instead of simulating energy minimization on continuous variables using a GPU, the TSU leverages the natural subthreshold thermal fluctuations of CMOS transistors. These fluctuations act as continuous noise that natively settles into discrete binary (Ising) states under applied voltages, physically solving the energy minimization problem.³⁹ This hardware alignment allows the Extropic architecture to execute Denoising Thermodynamic Models (DTMs)—a fusion of diffusion and energy-based models—up to 10,000 times more efficiently than state-of-the-art GPU equivalents.³⁹

Similarly, within the algorithmic domain of Boolean Satisfiability and combinatorial optimization, 2025–2026 frameworks such as Distributed Quantum Optimization (DQOF) demonstrate how

continuous-time quantum approximate optimization algorithms (QAOA) natively handle large-scale Higher-Order Unconstrained Binary Optimization (HUBO) problems.⁴² DQOF decomposes global continuous variables into sub-problems, allowing parallel quantum runs to collapse quantum superpositions into discrete binary strings. These sub-solutions are then deterministically aggregated back into discrete global assignments via a strict energy-improving update threshold.⁴² These hardware and physics-based paradigms provide compelling evidence that the optimal execution of the bounded continuous-to-Ising pipeline is achieved when the algorithmic constraints of the EBM perfectly align with the physical constraints of the executing medium, fundamentally bypassing the "Hardware Lottery" that restricts pure software approaches.³⁹

Architecture	Primary Citation	Claimed Transpilation Mechanism	Empirical Evidence & Scale	Open-Source
Extropic Thermodynamic Unit (TSU)	Freitas et al., 2024 ³⁹	Uses subthreshold CMOS thermal fluctuations to directly sample Ising distributions natively.	10,000x energy efficiency over GPU for Denoising Thermodynamic Models (DTMs).	No (Proprietary Hardware)
Distributed Quantum Optimization (DQOF)	Xu et al., arXiv:2604.20599 (2025) ⁴²	Quantum continuous evolutions are thresholded into discrete binary strings via energy updates.	Solves dense 500-variable HUBOs in 170s, outperforming standard QAOA solvers.	N/A

Honest Framing

The profound limitation of deploying Ising-substrate and thermodynamic hardware is the massive software redevelopment overhead required to interface with them. Standard continuous neural networks, transformers, and even traditional algorithmic EBMs cannot simply be compiled onto this hardware; they must be completely rewritten into Gibbs sampling logic or quadratic unconstrained binary forms.⁴⁰ Additionally, while physical systems naturally and perfectly handle the continuous-to-discrete collapse via annealing or voltage regulation,

modeling these dynamics analytically in software to interface with existing Phase-1 digital verifier ensembles introduces severe timing and state-space mismatches. Translating the exact physical collapse of an Ising spin network back into a differentiable software loss function for end-to-end training remains a mathematically intractable gap, restricting these thermodynamic systems primarily to isolated inference engines rather than deeply integrated foundation model training substrates.⁴⁰

Comparative Assessment of Transpilation Architectures

Modern Hopfield Networks	
PRIMARY METHOD	
Dense Associative Memory	
Gradient Preservation	-
Inference Latency	-
Resistance to Collapse	High
Hardware Alignment	High

Energy-Based Transformers (EBT)	
PRIMARY METHOD	
Energy Minimization	
Gradient Preservation	-
Inference Latency	High
Resistance to Collapse	High
Hardware Alignment	-

LittleBit	
PRIMARY METHOD	
Latent Matrix Factorization	
Gradient Preservation	High
Inference Latency	-
Resistance to Collapse	Medium
Hardware Alignment	-

Extropic TSU	
PRIMARY METHOD	
Denoising Thermodynamic Model	
Gradient Preservation	-
Inference Latency	Low
Resistance to Collapse	-
Hardware Alignment	High

Evaluation of continuous-to-Boolean transpilation architectures based on gradient preservation, inference latency, resistance to dimensional collapse, and alignment with neuromorphic/ising hardware.

Data sources: [Modern Hopfield Networks \(arXiv:2604.12811\)](#), [Energy-Based Transformers \(arXiv:2507.02092\)](#), [LittleBit \(NeurIPS 2025\)](#), [Extropic TSU \(ResearchGate\)](#)

3. Comparative Architecture Overview

Architecture Category	Primary Citation	Benchmark / Domain	Peer-Reviewed	Open Source	Scale	Claimed Transpilation Property	Known Failure Mode
Modern Hopfield (Continuous PDF)	Santos et al., 2025 ²	Dense Associative Mem.	Yes (ICLR 2025)	Yes	Millions	Continuous probability mass natively maps to discrete fixed points via CCCP optimization.	Sensitivity to inverse temperature limits; results in mixed continuous attractors without strict cooling. ¹
Dynamic EBM (EBT / NRGPT)	Gladstone 2025 ²⁷ / Dehmany 2025 ³⁰	System 2 Reasoning, NLP	Yes (arXiv preprint)	Yes	Up to 120B Tokens	Explicit gradient descent locates continuous minima for subsequent discrete token	High inference latency linearly tied to iterative gradient steps; requires intense gradient

						mappin g.	clampin g. ²⁷
Extrem e Boundi ng (LittleBi t / FSQ)	Lee et al., NeurIPS 2025 ¹⁶	Llama2 sub-1-b it QAT	Yes (NeurIP S)	Yes	13B Params	Dual-SV ID initialize s weights to preserv e gradien ts through strict sign(threshol d constrai nts.	Straight -Throug h Estimat ors (STE) fail if continu ous latents drift far from bounds, causing dead zones. ¹²
Stein Variati onal Gradie nt Descen t	He et al., 2024 / MDPI 2026 ²⁴	Model Predicti ve Control	Yes (MDPI)	Yes	Finite Horizon	Interact ing particle s minimiz e KL diverge nce in continu ous space before resolvin g to actions.	Kerneliz ed interacti ons scale poorly; repulsiv e forces can cause high-fre quency oscillati ons at Boolean borders ²³ .
Therm odyna	Freitas et al.,	Denoisi ng	Yes (PRX)	No	Edge / ASIC	Subthre shold	Severe softwar

mic Ising Hardw are (TSU)	2024 ³⁹	Thermo Models				thermal noise natively collaps es into discrete Ising energy states under voltage.	e-to-ha rdware mismat ch; cannot easily interfac e with existing continu ous softwar e logic. ⁴⁰
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4. Direct Answers to Target Inquiries

Q.A — Empirical Evidence of Information Preservation Under Sign-Discretization

The strongest empirical evidence confirming that sign-discretization of bounded continuous latents can preserve the precise information necessary for downstream symbolic reasoning stems from the 2025 NeurIPS paper introducing the **LittleBit** architecture (Lee et al.).¹⁶ LittleBit targets extreme LLM compression in the sub-1-bit regime (down to 0.1 bits per weight). Recognizing the catastrophic mutual information loss inherently associated with binarization (sign-discretization), the authors introduced the Dual Sign-Value-Independent Decomposition (Dual-SVID) for quantization-aware training. By implementing a multi-scale residual compensation mechanism that explicitly controls variance prior to the boundary, they proved mathematically and empirically that an Llama2-13B model could be aggressively bounded, strictly sign-discretized, and compressed to under 0.9 GB while simultaneously outperforming leading 0.7 BPW continuous-passthrough baseline techniques on perplexity benchmarks.¹⁶

Furthermore, parallel advancements in **Finite Scalar Quantization (FSQ)** applied to biological structure tokenization (Adaptive Protein Tokenization, 2026) and genomic expression foundation models (scPRINT-2, 2025) provide robust task performance comparisons.¹⁴ These studies prove that strictly bounding a latent space to $z \in [-1, 1]^d$ and quantizing across dimension-wise discrete levels (down to a binary split) perfectly preserves the topological reasoning required to predict high-fidelity multi-stage data.¹⁴ These frameworks confirm that, provided the pre-sign continuous representation applies factorized compensation to prevent distribution skew, the Boolean transpilation bottleneck retains the semantic density required for downstream verification tasks.

Q.B — Strongest Evidence of Transpilation Failure Modes

The literature repeatedly and consistently highlights that the continuous-to-Boolean transpilation pattern fails catastrophically under conditions of **parametric collapse** and **gradient death** when naive estimators are applied. Standard sign-discretization relies heavily on Straight-Through Estimators (STE) to artificially graft gradients backwards across the non-differentiable boundary. However, as documented in 2025 comprehensive reviews of vector and scalar quantization architectures, this approximation leads directly to *representation degeneracy* (the scalar equivalent of codebook collapse) if the continuous upstream variables lose their variance.¹² Specifically, if a continuous latent variable is pushed deeply into the bounds (e.g., $z = 0.999$), the STE will falsely report a gradient that attempts to adjust the upstream weights, but the magnitude of the resulting update will be insufficient to cross the zero-threshold. Consequently, the Boolean output remains completely locked, permanently freezing downstream symbolic action selection.¹²

Additionally, in dual-embedding energy-based models and dynamic physical systems, the continuous-to-discrete interface is highly susceptible to "critical collapse instability." Studies on non-equilibrium dynamics and parametric collapse indicate that continuous energy landscapes mapping to discrete multi-hump solutions exhibit severe vulnerabilities. If the interaction variables within the energy function are not strictly and properly normalized, the entire system degenerates into a uniform state that obliterates symbolic differentiation, causing all continuous latents to collapse into a single degenerate Boolean mode regardless of the input stimulus.⁴⁵

Q.C — Theoretical Optimality and Complexity

There is no proven single global optimal algorithm with polynomial complexity for the specific transpilation pattern $z \in [-1, 1]^d \rightarrow \text{sign}(z) \rightarrow \text{Ising}$. This is due to the fundamental, proven NP-hardness of optimally mapping continuous ambient spaces to exact Boolean partitions without arbitrary loss.⁴⁷ However, there are significant theoretical advances. Fixed-parameter tractable algorithms derived from recursive dynamic programming provide optimal theoretical bounds for proper decision trees mapping continuous distributions to Boolean spaces, provided that the depth and scale are strictly constrained.⁴⁷ Furthermore, 2025 extensions to the Kane-Tao Theorem provide explicit lower bounds for sum entropy when transitioning from continuous probability mass to the rigid Boolean cube $\{0, 1\}^d$, formally establishing the absolute limits of mutual information loss during sign-discretization.⁴⁸

Because exact global optimality is computationally NP-hard, the overwhelming consensus heuristic in current architectures is the deployment of boundary-restricted latents paired with Straight-Through Estimators (STE), but crucially augmented with quantization-aware initialization (e.g., Dual-SVID) or Gumbel-Softmax relaxations.¹² As identified in Q.B, the known failure mode of this heuristic is the vanishing gradient and frozen node problem, where the

surrogate gradient fails to capture the true distance required to cross the Boolean threshold.

Q.D — Peer-Reviewed "Sign-then-Ising" Pipelines at Scale

While architectures explicitly branded as "Sign-then-Ising" are niche, the functional equivalent pipeline operates at scale in several highly prominent 2024–2026 production and research-grade systems:

1. **LittleBit (NeurIPS 2025)**: Represents a production-grade application of this pipeline, binarizing low-rank matrix factors mapping continuous LLM weights directly into Boolean representations at massive scale (Llama2 13B) to achieve extreme compression.¹⁶
2. **scPRINT-2 (FSQ-VAE, 2025)**: A research-grade foundation model in genomics that explicitly enforces continuous dimension boundaries $z \in [-1, 1]$ before dimension-wise discrete level quantization to map continuous gene expression states into distinct logical pathways.¹⁴
3. **Extropic Denoising Thermodynamic Models (DTM, PRX 2024)**: While executing natively in hardware rather than software, this represents the purest physical expression of the pipeline. It maps continuous probability distributions into a subthreshold CMOS structure that yields explicit, discrete Ising spins for probabilistic inference.³⁹
4. **Distributed Quantum Optimization Framework (DQOF) (arXiv:2604.20599, 2025)**: Implements continuous quantum spatial evolutions that are mathematically thresholded into discrete binary strings via strict energy-improving logic to heuristically solve massive 500-variable Higher-Order Unconstrained Binary Optimization (HUBO) systems.⁴²

Q.E — Consensus on Substrate Choice for Energy-Based Foundation Models

The 2025–2026 consensus strongly favors the **bounded-continuous-to-Boolean (or low-cardinality FSQ) transpilation pipeline** over purely continuous or purely discrete architectures.

- *Full Continuous (No Discretization)*: While continuous-time EBTs, SVGD models, and traditional diffusion frameworks possess immense representational fluidity, they suffer from unbounded error accumulation during long-horizon iterative rollouts and fail entirely to interface cleanly with discrete logical verifiers (Phase-1 SAT solvers) without collapsing the constraint logic.¹⁹
- *Full Discrete (Token-Only)*: Traditional discrete LLMs and strict VQ-VAEs are increasingly abandoned for continuous-latent applications due to computational rigidity and severe codebook collapse, rendering them unable to allocate continuous neural resources proportionally to dynamic task complexity.⁴
- *The Bounded-Discrete Synthesis*: The literature decisively converges on bounded-latent scalar quantization. As demonstrated by the successes of FSQ and LittleBit, compressing representation into mathematically rigid boundaries before Boolean/scalar mapping completely eliminates the need for expensive codebooks. Crucially, it preserves the

continuous fluid gradient search (System-2 energy descent thinking) right up to the final moment prior to the commitment of a discrete symbolic action.¹³ Carnot's commitment to this specific pipeline is entirely validated by the explicit trajectory of frontier 2026 foundation model research.

5. Gaps and Recommendations

Priority Reading

To fortify the Carnot architecture's mathematical and structural foundations, immediate attention should be directed toward the following advanced 2025/2026 mechanics:

- **Dual Sign-Value-Independent Decomposition (Dual-SVID)** (Lee et al., *NeurIPS* 2025)¹⁶: Essential for understanding how to mathematically preserve gradient flow upstream of the $\text{sign}(z)$ operation. It provides the exact blueprint for initializing variance without relying entirely on naive STE heuristics.
- **Energy-Based Transformers (EBT)** (Gladstone et al., *arXiv:2507.02092*)²⁷: Priority reading for structuring the Phase-3 continuous latent reasoning. The EBT methodology for executing inference via continuous gradient descent on an energy verifier perfectly mirrors the required continuous build-up prior to Carnot's Ising discretization.
- **NRGPT and Continuous MHN Dynamics** (Dehmamy et al., 2025)¹: Provides the necessary theoretical baseline for treating continuous tokens as particles interacting on an energy landscape, specifically addressing the interaction rates and energy formulation needed to ensure asymptotic stability across the manifold.

Architectures for Direct Comparison

Carnot must strategically position itself against **Finite Scalar Quantization (FSQ)** based transformers and **Denoising Thermodynamic Models (DTM)**. FSQ represents the absolute software industry standard for bounded-latent discretization; clearly establishing how Carnot's strict 1-bit Boolean/Ising transpilation offers computational, energetic, or logical grounding advantages over multi-level FSQ is vital.¹² Conversely, Extropic's TSU establishes the ultimate hardware baseline. Carnot should be conceptually framed as the algorithmic software bridge that provides the same efficient energy-landscape Ising logic as the TSU, but without the dependency on highly specialized, proprietary thermodynamic CMOS hardware.³⁹

Novelty Boundaries (What NOT to Claim)

Do not claim that the "Sign-then-Ising" transpilation natively eliminates gradient vanishing or that Straight-Through Estimators (STE) are a solved problem inherent to this architecture. The literature extensively proves that strict sign-discretization is fundamentally destructive and heavily prone to representation lock-in without the addition of sophisticated upstream variance compensation.¹² Furthermore, do not claim that energy-based continuous token exploration is a purely novel concept invented by the project; models like EBT, ET, and NRGPT have firmly

established continuous energy-descent inference throughout 2025.²⁷ Carnot's specific, defensible novelty lies strictly in utilizing this transpilation pipeline as the *load-bearing path between continuous latent reasoning and discrete neuro-symbolic verifier ensembles*, rather than merely utilizing it as a tool for model compression or general generation.

Labs and Researchers to Engage

- **Alexi Gladstone & Tariq Iqbal (UVA/UIUC)**: Lead authors on the Energy-Based Transformers (EBT) research. Their profound insights into scaling EBM for continuous System-2 thinking align directly with the required mechanics of Carnot's Phase-3 reasoning layer.²⁷
- **Dmitry Krotov & Jean-Jacques Slotine (IBM/MIT)**: Pioneers in Modern Hopfield Networks and NRGPT. Their extensive work on dense associative memories and dynamic continuous energy states is foundational to navigating the continuous-discrete energy interface.³⁰
- **Trevor McCourt & Guillaume Verdon (Extropic)**: The primary architects behind thermodynamic Ising hardware. Engaging their research output could validate how Carnot's software-based substrate directly aligns with the operational logic of next-generation physical hardware deployment.³⁹

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